Complete Kinematic Analysis Report

Auto-Generated

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Contents

1 Transformation Matrices

Matrix T_1

$$\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & L\cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & L\sin(\theta_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_2

$$\begin{bmatrix} \cos(\theta_2) & 0 & -\sin(\theta_2) & 2L\cos(\theta_2) \\ \sin(\theta_2) & 0 & \cos(\theta_2) & 2L\sin(\theta_2) \\ 0 & -1 & 0 & 5L \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_3

$$\begin{bmatrix} -\cos(\theta_3) & 0 & -\sin(\theta_3) & 0 \\ -\sin(\theta_3) & 0 & \cos(\theta_3) & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_4

$$\begin{bmatrix} \cos(\theta_4) & 0 & -\sin(\theta_4) & \frac{5L\cos(\theta_4)}{2} \\ \sin(\theta_4) & 0 & \cos(\theta_4) & \frac{5L\sin(\theta_4)}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_5

$$\begin{bmatrix} \cos(\theta_5) & 0 & \sin(\theta_5) & \frac{5L\cos(\theta_5)}{2} \\ \sin(\theta_5) & 0 & -\cos(\theta_5) & \frac{5L\sin(\theta_5)}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_6

$$\begin{bmatrix} -\cos(\theta_6) & 0 & \sin(\theta_6) & 0 \\ -\sin(\theta_6) & 0 & -\cos(\theta_6) & 0 \\ 0 & -1 & 0 & \frac{5L}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_7

$$\begin{bmatrix} \cos(\theta_7) & 0 & -\sin(\theta_7) & 2L\cos(\theta_7) \\ \sin(\theta_7) & 0 & \cos(\theta_7) & 2L\sin(\theta_7) \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 Loop Closure Equations

2.1 Set 1: $T_1T_2T_3T_4 = T_7^{-1}T_6^{-1}T_5^{-1}$

2.2 Set 2: $T_2T_3T_4T_5 = T_1^{-1}T_7^{-1}T_6^{-1}$

2.3 Set 3: $T_3T_4T_5T_6 = T_2^{-1}T_1^{-1}T_7^{-1}$

2.4 Set 4: $T_4T_5T_6T_7 = T_3^{-1}T_2^{-1}T_1^{-1}$

2.5 Set 5: $T_5T_6T_7T_1 = T_4^{-1}T_3^{-1}T_2^{-1}$

2.6 Set 6: $T_6T_7T_1T_2 = T_5^{-1}T_4^{-1}T_3^{-1}$

2.7 Set 7: $T_7T_1T_2T_3 = T_6^{-1}T_5^{-1}T_4^{-1}$

3 Simplified Scalar Equations

3.1 Set 1: $T_1T_2T_3T_4 = T_7^{-1}T_6^{-1}T_5^{-1}$

Permutation 1, (a_x)

$$-\sin(\theta_1)\sin(\theta_2)\sin(\theta_4)\cos(\theta_3) - \sin(\theta_1)\cos(\theta_2)\cos(\theta_4) - \sin(\theta_2)\cos(\theta_1)\cos(\theta_4) + \sin(\theta_4)\cos(\theta_1)\cos(\theta_2)\cos(\theta_3) - \frac{\sin(\theta_6)\sin^2(\theta_7)}{\cos(\theta_7)} + \frac{\sin(\theta_6)}{\cos(\theta_7)} = 0$$

Permutation 1, (p_x)

$$\frac{L\left(-5\sin(\theta_4)\sin(\theta_1+\theta_2) - 5\sin(\theta_7) + 2\cos(\theta_1) - 5\cos(\theta_3)\cos(\theta_4)\cos(\theta_1+\theta_2) - 5\cos(\theta_6)\cos(\theta_7) + 4\cos\right.}{2} = 0$$

Permutation 1, (n_y)

$$-\sin(\theta_1)\sin(\theta_2)\sin(\theta_4) - \sin(\theta_1)\cos(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_2)\cos(\theta_1)\cos(\theta_3)\cos(\theta_4) + \sin(\theta_4)\cos(\theta_1)\cos(\theta_2) - \frac{\sin^2(\theta_5)\sin(\theta_6)}{\cos(\theta_5)} + \frac{\sin(\theta_6)}{\cos(\theta_5)} = 0$$

Permutation 1, (o_y)

$$\sin(\theta_1)\sin(\theta_3)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6) = 0$$

Permutation 1, (a_y)

$$-\sin(\theta_1)\sin(\theta_2)\cos(\theta_4) + \sin(\theta_1)\sin(\theta_4)\cos(\theta_2)\cos(\theta_3) + \sin(\theta_2)\sin(\theta_4)\cos(\theta_1)\cos(\theta_3) + \cos(\theta_1)\cos(\theta_2)\cos(\theta_4) - \cos(\theta_6) = 0$$

Permutation 1, (p_y)

$$\frac{L\left(2\sin(\theta_1) + 5\sin(\theta_4)\cos(\theta_1+\theta_2) - 5\sin(\theta_6) - 5\sin(\theta_1+\theta_2)\cos(\theta_3)\cos(\theta_4) + 4\sin(\theta_1+\theta_2)\right)}{2} = 0$$

Permutation 1, (n_z)

$$\sin(\theta_3)\cos(\theta_4) - \frac{\sin^2(\theta_5)\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_5)\cos(\theta_6)} + \frac{\sin^2(\theta_5)\sin(\theta_7)}{\cos(\theta_5)\cos(\theta_6)} + \sin(\theta_5)\cos(\theta_7) + \frac{\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_5)\cos(\theta_6)} - \frac{\sin(\theta_7)}{\cos(\theta_5)\cos(\theta_6)} = 0$$

Permutation 1, (o_z)

$$\frac{\sin(\theta_5)\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_6)} - \frac{\sin(\theta_5)\sin(\theta_7)}{\cos(\theta_6)} + \cos(\theta_3) - \cos(\theta_5)\cos(\theta_7) = 0$$

Permutation 1, (a_z)

$$-\sin(\theta_3)\sin(\theta_4) - \sin(\theta_6)\sin(\theta_7) = 0$$

Permutation 1, (p_z)

$$\frac{5L\left(\sin(\theta_3)\cos(\theta_4) + \sin(\theta_7)\cos(\theta_6) - \cos(\theta_7) + 2\right)}{2} = 0$$

3.2 Set 2: $T_2T_3T_4T_5 = T_1^{-1}T_7^{-1}T_6^{-1}$

Permutation 2, (o_x)

$$\frac{\sin^2(\theta_1)\sin(\theta_6)\sin^2(\theta_7)}{\cos(\theta_1)\cos(\theta_7)} - \frac{\sin^2(\theta_1)\sin(\theta_6)}{\cos(\theta_1)\cos(\theta_7)} - \sin(\theta_1)\cos(\theta_6) - \sin(\theta_2)\cos(\theta_4) + \sin(\theta_4)\cos(\theta_2)\cos(\theta_3) - \frac{\sin(\theta_6)\sin^2(\theta_7)}{\cos(\theta_1)\cos(\theta_7)} + \frac{\sin(\theta_6)}{\cos(\theta_1)\cos(\theta_7)} = 0$$

Permutation 2, (a_x)

$$-\frac{\sin^2(\theta_1)\sin(\theta_7)}{\cos(\theta_1)} - \sin(\theta_2)\sin(\theta_4)\sin(\theta_5) - \sin(\theta_3)\cos(\theta_2)\cos(\theta_5) - \sin(\theta_5)\cos(\theta_2)\cos(\theta_3)\cos(\theta_4) + \frac{\sin(\theta_7)}{\cos(\theta_1)} = 0$$

Permutation 2, (p_x)

$$\frac{L\left(-5\sin(\theta_2)\sin(\theta_4)\cos(\theta_5) - 5\sin(\theta_2)\sin(\theta_4) + 5\sin(\theta_3)\sin(\theta_5)\cos(\theta_2) - 5\sin(\theta_7)\cos(\theta_1) + 4\cos(\theta_1) -\right.}{2}$$

$$= 0$$

Permutation 2, (o_y)

$$\frac{\sin(\theta_1)\sin(\theta_6)\sin^2(\theta_7)}{\cos(\theta_7)} - \frac{\sin(\theta_1)\sin(\theta_6)}{\cos(\theta_7)} + \sin(\theta_2)\sin(\theta_4)\cos(\theta_3) - \cos(\theta_1)\cos(\theta_6) + \cos(\theta_2)\cos(\theta_4) = 0$$

Permutation 2, (a_y)

$$-\sin(\theta_1)\sin(\theta_7) - \sin(\theta_2)\sin(\theta_3)\cos(\theta_5) \\ -\sin(\theta_2)\sin(\theta_5)\cos(\theta_3)\cos(\theta_4) + \sin(\theta_4)\sin(\theta_5)\cos(\theta_2) = 0$$

Permutation 2, (p_y)

$$\frac{L\left(-5\left(\sin(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2)\right)\cos(\theta_5) + 5\sin(\theta_1)\sin(\theta_7) - 4\sin(\theta_1) + 5\sin(\theta_2)\sin(\theta_3)\right.}{2} \\ = 0$$

Permutation 2, (n_z)

$$\sin(\theta_3)\cos(\theta_4)\cos(\theta_5) + \sin(\theta_5)\cos(\theta_3) + \frac{\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_6)} - \frac{\sin(\theta_7)}{\cos(\theta_6)} = 0$$

Permutation 2, (o_z)

$$-\sin(\theta_3)\sin(\theta_4) - \sin(\theta_6)\sin(\theta_7) = 0$$

Permutation 2, (a_z)

$$\sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5) + \cos(\theta_7) = 0$$

Permutation 2, (p_z)

$$\frac{5L\left(\sin(\theta_3)\cos(\theta_4)\cos(\theta_5) + \sin(\theta_3)\cos(\theta_4) + \sin(\theta_5)\cos(\theta_3) - \cos(\theta_7) + 2\right)}{2} = 0$$

3.3 Set 3: $T_3T_4T_5T_6 = T_2^{-1}T_1^{-1}T_7^{-1}$

Permutation 3, (a_x)

$$-\frac{\sin(\theta_1)\sin^2(\theta_2)}{\cos(\theta_2)} + \frac{\sin(\theta_1)}{\cos(\theta_2)} + \sin(\theta_2)\cos(\theta_1) + \sin(\theta_3)\sin(\theta_5)\sin(\theta_6) \\ -\sin(\theta_4)\cos(\theta_3)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_3)\cos(\theta_4)\cos(\theta_5) = 0$$

Permutation 3, (p_x)

$$\frac{L\left(-4\sin(\theta_1)\sin(\theta_2) - 5\sqrt{2}\sin(\theta_3)\cos\left(\theta_5 + \frac{\pi}{4}\right) - 5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_3)\cos(\theta_4) + 4\cos(\theta_1)\cos(\theta_2) + 2\right.}{2} \\ = 0$$

Permutation 3, (n_y)

$$-\sin(\theta_3)\sin(\theta_4)\sin(\theta_6) + \sin(\theta_3)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6) + \sin(\theta_5)\cos(\theta_3)\cos(\theta_6) - \sin(\theta_7) = 0$$

Permutation 3, (o_y)

$$\sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5) + \cos(\theta_7) = 0$$

Permutation 3, (a_y)

$$-\sin(\theta_3)\sin(\theta_4)\cos(\theta_6)-\sin(\theta_3)\sin(\theta_6)\cos(\theta_4)\cos(\theta_5)-\sin(\theta_5)\sin(\theta_6)\cos(\theta_3)=0$$

Permutation 3, (p_y)

$$\frac{5L\left(-\sqrt{2}\sin(\theta_3)\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_4)-\sin(\theta_3)\cos(\theta_4)+\sqrt{2}\cos(\theta_3)\cos\left(\theta_5+\frac{\pi}{4}\right)-2\right)}{2}=0$$

Permutation 3, (o_z)

$$-\frac{\sin^2(\theta_1)\sin(\theta_2)\sin(\theta_7)}{\cos(\theta_1)}+\sin(\theta_1)\sin(\theta_7)\cos(\theta_2)+\frac{\sin(\theta_2)\sin(\theta_7)}{\cos(\theta_1)}-\sin(\theta_4)\sin(\theta_5)=0$$

Permutation 3, (a_z)

$$-\sin(\theta_1)\sin(\theta_2)+\sin(\theta_4)\sin(\theta_6)\cos(\theta_5)+\cos(\theta_1)\cos(\theta_2)-\cos(\theta_4)\cos(\theta_6)=0$$

Permutation 3, (p_z)

$$\frac{L\left(-4\sin(\theta_1)\cos(\theta_2)-4\sin(\theta_2)\cos(\theta_1)-2\sin(\theta_2)+5\sqrt{2}\sin(\theta_4)\sin\left(\theta_5+\frac{\pi}{4}\right)+5\sin(\theta_4)\right)}{2}=0$$

3.4 Set 4: $T_4T_5T_6T_7 = T_3^{-1}T_2^{-1}T_1^{-1}$

Permutation 4, (a_x)

$$-\sin(\theta_3)-\sin(\theta_4)\sin(\theta_6)\sin(\theta_7)-\sin(\theta_5)\cos(\theta_4)\cos(\theta_7)+\sin(\theta_7)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)=0$$

Permutation 4, (p_x)

$$\frac{L\left(10\sin(\theta_3)+4\sin(\theta_4)\sin(\theta_6)\cos(\theta_7)-4\sin(\theta_5)\sin(\theta_7)\cos(\theta_4)+5\sqrt{2}\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_4)-2\cos(\theta_2)\mathrm{c}\right.}{2}$$

$$=0$$

Permutation 4, (n_y)

$$-\frac{\sin^2(\theta_1)\sin(\theta_2)}{\cos(\theta_1)}+\sin(\theta_1)\cos(\theta_2)+\frac{\sin(\theta_2)}{\cos(\theta_1)}-\sin(\theta_4)\sin(\theta_5)\sin(\theta_7)$$
$$-\sin(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)-\sin(\theta_6)\cos(\theta_4)\cos(\theta_7)=0$$

Permutation 4, (o_y)

$$\sin(\theta_1)\sin(\theta_2)-\sin(\theta_4)\sin(\theta_6)\cos(\theta_5)-\cos(\theta_1)\cos(\theta_2)+\cos(\theta_4)\cos(\theta_6)=0$$

Permutation 4, (a_y)

$$-\sin(\theta_4)\sin(\theta_5)\cos(\theta_7)+\sin(\theta_4)\sin(\theta_7)\cos(\theta_5)\cos(\theta_6)+\sin(\theta_6)\sin(\theta_7)\cos(\theta_4)=0$$

Permutation 4, (p_y)

$$\frac{L\left(-2\sin\left(\theta_2\right)-4\sin\left(\theta_4\right)\sin\left(\theta_5\right)\sin\left(\theta_7\right)+5\sqrt{2}\sin\left(\theta_4\right)\sin\left(\theta_5+\frac{\pi}{4}\right)-4\sin\left(\theta_4\right)\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)+5\,\mathrm{s}\right.}{2}$$

$$=0$$

Permutation 4, (o_z)

$$-\frac{\sin\left(\theta_1\right)\sin^2\left(\theta_2\right)\sin\left(\theta_3\right)}{\cos\left(\theta_2\right)}+\frac{\sin\left(\theta_1\right)\sin\left(\theta_3\right)}{\cos\left(\theta_2\right)}+\sin\left(\theta_2\right)\sin\left(\theta_3\right)\cos\left(\theta_1\right)+\sin\left(\theta_5\right)\sin\left(\theta_6\right)=0$$

Permutation 4, (a_z)

$$-\sin\left(\theta_5\right)\sin\left(\theta_7\right)\cos\left(\theta_6\right)+\cos\left(\theta_3\right)-\cos\left(\theta_5\right)\cos\left(\theta_7\right)=0$$

Permutation 4, (p_z)

$$\frac{L\left(-2\sin\left(\theta_3\right)\cos\left(\theta_2\right)-4\sin\left(\theta_3\right)+4\sin\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)-4\sin\left(\theta_7\right)\cos\left(\theta_5\right)-10\cos\left(\theta_3\right)+5\sqrt{2}\cos\left(\theta_5\right.\right.}{2}$$

$$=0$$

3.5 Set 5: $T_5T_6T_7T_1 = T_4^{-1}T_3^{-1}T_2^{-1}$

Permutation 5, (a_x)

$$\frac{\sin\left(\theta_3\right)\sin^2\left(\theta_4\right)}{\cos\left(\theta_4\right)}-\frac{\sin\left(\theta_3\right)}{\cos\left(\theta_4\right)}-\sin\left(\theta_5\right)\cos\left(\theta_7\right)+\sin\left(\theta_7\right)\cos\left(\theta_5\right)\cos\left(\theta_6\right)=0$$

Permutation 5, (p_x)

$$\frac{L\left(-2\sin\left(\theta_1\right)\sin\left(\theta_6\right)\cos\left(\theta_5\right)+10\sin\left(\theta_3\right)\cos\left(\theta_4\right)-2\sin\left(\theta_5\right)\sin\left(\theta_7\right)\cos\left(\theta_1\right)-4\sin\left(\theta_5\right)\sin\left(\theta_7\right)+5\sqrt{2}\sin\left(\right.\right.}{2}$$

$$=0$$

Permutation 5, (n_y)

$$\begin{aligned}&-\sin\left(\theta_1\right)\sin\left(\theta_5\right)\sin\left(\theta_6\right)+\frac{\sin^2\left(\theta_2\right)\sin\left(\theta_3\right)}{\cos\left(\theta_2\right)}-\frac{\sin\left(\theta_3\right)}{\cos\left(\theta_2\right)}\\&-\sin\left(\theta_5\right)\cos\left(\theta_1\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)+\sin\left(\theta_7\right)\cos\left(\theta_1\right)\cos\left(\theta_5\right)=0\end{aligned}$$

Permutation 5, (o_y)

$$\begin{aligned}&\sin\left(\theta_1\right)\sin\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)-\sin\left(\theta_1\right)\sin\left(\theta_7\right)\cos\left(\theta_5\right)\\&-\sin\left(\theta_2\right)\sin\left(\theta_3\right)-\sin\left(\theta_5\right)\sin\left(\theta_6\right)\cos\left(\theta_1\right)=0\end{aligned}$$

Permutation 5, (a_y)

$$\sin\left(\theta_5\right)\sin\left(\theta_7\right)\cos\left(\theta_6\right)-\cos\left(\theta_3\right)+\cos\left(\theta_5\right)\cos\left(\theta_7\right)=0$$

Permutation 5, (p_y)

$$\frac{L\left(-2\left(\sin\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)-\sin\left(\theta_7\right)\cos\left(\theta_5\right)\right)\cos\left(\theta_1\right)-2\sin\left(\theta_1\right)\sin\left(\theta_5\right)\sin\left(\theta_6\right)+4\sin\left(\theta_3\right)-4\sin\left(\theta_5\right)\right.}{2}$$

$$= 0$$

Permutation 5, (n_z)

$$\sin\left(\theta_1\right)\cos\left(\theta_6\right)-\frac{\sin^2\left(\theta_2\right)\sin^2\left(\theta_3\right)\sin\left(\theta_4\right)}{\cos\left(\theta_2\right)\cos\left(\theta_3\right)}+\frac{\sin^2\left(\theta_2\right)\sin\left(\theta_4\right)}{\cos\left(\theta_2\right)\cos\left(\theta_3\right)}+\sin\left(\theta_2\right)\cos\left(\theta_4\right)$$
$$+\frac{\sin^2\left(\theta_3\right)\sin\left(\theta_4\right)}{\cos\left(\theta_2\right)\cos\left(\theta_3\right)}-\frac{\sin\left(\theta_4\right)}{\cos\left(\theta_2\right)\cos\left(\theta_3\right)}-\sin\left(\theta_6\right)\cos\left(\theta_1\right)\cos\left(\theta_7\right)=0$$

Permutation 5, (o_z)

$$\sin\left(\theta_1\right)\sin\left(\theta_6\right)\cos\left(\theta_7\right)+\frac{\sin\left(\theta_2\right)\sin^2\left(\theta_3\right)\sin\left(\theta_4\right)}{\cos\left(\theta_3\right)}$$
$$-\frac{\sin\left(\theta_2\right)\sin\left(\theta_4\right)}{\cos\left(\theta_3\right)}+\cos\left(\theta_1\right)\cos\left(\theta_6\right)-\cos\left(\theta_2\right)\cos\left(\theta_4\right)=0$$

Permutation 5, (a_z)

$$\sin\left(\theta_3\right)\sin\left(\theta_4\right)+\sin\left(\theta_6\right)\sin\left(\theta_7\right)=0$$

Permutation 5, (p_z)

$$L\left(\sin\left(\theta_1\right)\cos\left(\theta_6\right)-5\sin\left(\theta_3\right)\sin\left(\theta_4\right)+2\sin\left(\theta_4\right)\cos\left(\theta_3\right)\right.$$
$$\left.-\sin\left(\theta_6\right)\cos\left(\theta_1\right)\cos\left(\theta_7\right)-2\sin\left(\theta_6\right)\cos\left(\theta_7\right)\right)=0$$

3.6 Set 6: $T_6T_7T_1T_2 = T_5^{-1}T_4^{-1}T_3^{-1}$

Permutation 6, (o_x)

$$\frac{\sin\left(\theta_3\right)\sin^2\left(\theta_4\right)\sin^2\left(\theta_5\right)}{\cos\left(\theta_4\right)\cos\left(\theta_5\right)}-\frac{\sin\left(\theta_3\right)\sin^2\left(\theta_4\right)}{\cos\left(\theta_4\right)\cos\left(\theta_5\right)}-\frac{\sin\left(\theta_3\right)\sin^2\left(\theta_5\right)}{\cos\left(\theta_4\right)\cos\left(\theta_5\right)}$$
$$+\frac{\sin\left(\theta_3\right)}{\cos\left(\theta_4\right)\cos\left(\theta_5\right)}+\sin\left(\theta_5\right)\cos\left(\theta_3\right)-\sin\left(\theta_7\right)\cos\left(\theta_6\right)=0$$

Permutation 6, (a_x)

$$\sin\left(\theta_1\right)\sin\left(\theta_2\right)\sin\left(\theta_6\right)+\sin\left(\theta_1\right)\cos\left(\theta_2\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)+\sin\left(\theta_2\right)\cos\left(\theta_1\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)$$
$$+\frac{\sin\left(\theta_4\right)\sin^2\left(\theta_5\right)}{\cos\left(\theta_5\right)}-\frac{\sin\left(\theta_4\right)}{\cos\left(\theta_5\right)}-\sin\left(\theta_6\right)\cos\left(\theta_1\right)\cos\left(\theta_2\right)=0$$

Permutation 6, (p_x)

$$\frac{L\left(-2\sin\left(\theta_1\right)\sin\left(\theta_6\right)-4\sin\left(\theta_6\right)\sin\left(\theta_1+\theta_2\right)+10\sin\left(\theta_7\right)\cos\left(\theta_6\right)-2\cos\left(\theta_1\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right)+5\cos\left(\theta_5\right)-\right.}{2}$$

$$= 0$$

Permutation 6, (n_y)

$$\sin(\theta_1)\sin(\theta_2)\sin(\theta_6)\cos(\theta_7)+\sin(\theta_1)\cos(\theta_2)\cos(\theta_6)+\sin(\theta_2)\cos(\theta_1)\cos(\theta_6)$$
$$+\frac{\sin^2(\theta_3)\sin(\theta_4)}{\cos(\theta_3)}-\frac{\sin(\theta_4)}{\cos(\theta_3)}-\sin(\theta_6)\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)=0$$

Permutation 6, (o_y)

$$-\sin(\theta_3)\sin(\theta_4)-\sin(\theta_6)\sin(\theta_7)=0$$

Permutation 6, (a_y)

$$-\sin(\theta_1)\sin(\theta_2)\cos(\theta_6)+\sin(\theta_1)\sin(\theta_6)\cos(\theta_2)\cos(\theta_7)$$
$$+\sin(\theta_2)\sin(\theta_6)\cos(\theta_1)\cos(\theta_7)+\cos(\theta_1)\cos(\theta_2)\cos(\theta_6)-\cos(\theta_4)=0$$

Permutation 6, (p_y)

$$L(\sin(\theta_1)\cos(\theta_6)+5\sin(\theta_6)\sin(\theta_7)-\sin(\theta_6)\cos(\theta_1)\cos(\theta_7)$$
$$-2\sin(\theta_6)\cos(\theta_7)\cos(\theta_1+\theta_2)-2\sin(\theta_6)\cos(\theta_7)+2\sin(\theta_1+\theta_2)\cos(\theta_6))=0$$

Permutation 6, (o_z)

$$-\frac{\sin(\theta_3)\sin^2(\theta_4)\sin(\theta_5)}{\cos(\theta_4)}+\frac{\sin(\theta_3)\sin(\theta_5)}{\cos(\theta_4)}-\cos(\theta_3)\cos(\theta_5)+\cos(\theta_7)=0$$

Permutation 6, (a_z)

$$\sin(\theta_1)\sin(\theta_7)\cos(\theta_2)+\sin(\theta_2)\sin(\theta_7)\cos(\theta_1)-\sin(\theta_4)\sin(\theta_5)=0$$

Permutation 6, (p_z)

$$\frac{L(5\sin(\theta_5)-2\sin(\theta_7)\cos(\theta_1)-4\sin(\theta_7)\cos(\theta_1+\theta_2)-4\sin(\theta_7)-10\cos(\theta_7)+5)}{2}=0$$

3.7 Set 7: $T_7T_1T_2T_3 = T_6^{-1}T_5^{-1}T_4^{-1}$

Permutation 7, (a_x)

$$\sin(\theta_1)\sin(\theta_2)\sin(\theta_3)\cos(\theta_7)-\sin(\theta_3)\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)$$
$$+\frac{\sin(\theta_5)\sin^2(\theta_6)}{\cos(\theta_6)}-\frac{\sin(\theta_5)}{\cos(\theta_6)}+\sin(\theta_7)\cos(\theta_3)=0$$

Permutation 7, (p_x)

$$\frac{L(-10\sin(\theta_7)+2\cos(\theta_1)\cos(\theta_7)-5\cos(\theta_5)\cos(\theta_6)-5\cos(\theta_6)+4\cos(\theta_7)\cos(\theta_1+\theta_2)+4\cos(\theta_7))}{2}$$
$$=0$$

Permutation 7, (n_y)

$$\sin(\theta_1)\sin(\theta_2)\sin(\theta_7)\cos(\theta_3) + \sin(\theta_3)\cos(\theta_7) - \frac{\sin^2(\theta_4)\sin(\theta_5)}{\cos(\theta_4)} + \frac{\sin(\theta_5)}{\cos(\theta_4)} - \sin(\theta_7)\cos(\theta_1)\cos(\theta_2)\cos(\theta_3) = 0$$

Permutation 7, (o_y)

$$-\sin(\theta_1)\sin(\theta_7)\cos(\theta_2) - \sin(\theta_2)\sin(\theta_7)\cos(\theta_1) + \sin(\theta_4)\sin(\theta_5) = 0$$

Permutation 7, (a_y)

$$\sin(\theta_1)\sin(\theta_2)\sin(\theta_3)\sin(\theta_7) - \sin(\theta_3)\sin(\theta_7)\cos(\theta_1)\cos(\theta_2) - \cos(\theta_3)\cos(\theta_7) + \cos(\theta_5) = 0$$

Permutation 7, (p_y)

$$\frac{L\left(-5\sin(\theta_5) + 2\sin(\theta_7)\cos(\theta_1) + 4\sin(\theta_7)\cos(\theta_1+\theta_2) + 4\sin(\theta_7) + 10\cos(\theta_7) - 5\right)}{2} = 0$$

Permutation 7, (o_z)

$$\sin(\theta_1)\sin(\theta_2) + \frac{\sin(\theta_4)\sin^2(\theta_5)\sin(\theta_6)}{\cos(\theta_5)} - \frac{\sin(\theta_4)\sin(\theta_6)}{\cos(\theta_5)} - \cos(\theta_1)\cos(\theta_2) + \cos(\theta_4)\cos(\theta_6) = 0$$

Permutation 7, (a_z)

$$\sin(\theta_1)\sin(\theta_3)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6) = 0$$

Permutation 7, (p_z)

$$\frac{L\left(-2\sin(\theta_1) + 5\sin(\theta_6)\cos(\theta_5) + 5\sin(\theta_6) - 4\sin(\theta_1+\theta_2)\right)}{2} = 0$$

4 Single Driver Solutions

No complete solution found for single drivers.

5 Dual Driver Solutions

5.1 Inputs θ_3, θ_5

1. **Solve for** θ_4 using Permutation 3, (p_y)
   Equation form: $A\sin(\theta_4) + B\cos(\theta_4) + C = 0$

$$A = 0$$

$$B = -\frac{5L\left(\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right) + 1\right)\sin(\theta_3)}{2}$$

$$C = \frac{5L\left(\sqrt{2}\cos(\theta_3)\cos\left(\theta_5 + \frac{\pi}{4}\right) - 2\right)}{2}$$

Solution (via $t = \tan\frac{\theta_4}{2}$):

$$\theta_4 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

2. **Solve for** θ_7 using Permutation 2, (a_z)
Equation form: $A\sin(\theta_7) + B\cos(\theta_7) + C = 0$

$$\begin{aligned}
&A = 0\\
&B = 1\\
&C = \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5)
\end{aligned}$$

Solution (via $t = \tan\frac{\theta_7}{2}$):

$$\theta_7 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

3. **Solve for** θ_6 using Permutation 1, (n_z)
Equation form: $A\sin(\theta_6) + B\cos(\theta_6) + C = 0$

$$\begin{aligned}
&A = 0\\
&B = 0\\
&C = \sin(\theta_3)\cos(\theta_4) + \sin(\theta_5)\cos(\theta_7) + \tilde{\infty}\sin(\theta_7)\cos(\theta_5)
\end{aligned}$$

Solution (via $t = \tan\frac{\theta_6}{2}$):

$$\theta_6 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

4. **Solve for** θ_2 using Permutation 4, (p_x)
Equation form: $A\sin(\theta_2) + B\cos(\theta_2) + C = 0$

$$\begin{aligned}
&A = 0\\
&B = -L\cos(\theta_3)\\
&C\\
&= \frac{L\left(10\sin(\theta_3) + 4\sin(\theta_4)\sin(\theta_6)\cos(\theta_7) - 4\sin(\theta_5)\sin(\theta_7)\cos(\theta_4) + 5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_4) - 4\cos(\theta_3) - 4\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) + \right.}{2}
\end{aligned}$$

Solution (via $t = \tan\frac{\theta_2}{2}$):

$$\theta_2 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

5. **Solve for** θ_1 using Permutation 1, (n_x)
Equation form: $A\sin(\theta_1) + B\cos(\theta_1) + C = 0$

$$\begin{aligned}
&A = \sin(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2)\\
&B = -\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4)\\
&C = \sin(\theta_5)\sin(\theta_7) + \cos(\theta_5)\cos(\theta_6)\cos(\theta_7)
\end{aligned}$$

Solution (via $t = \tan\frac{\theta_1}{2}$):

$$\theta_1 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

5.2 Inputs θ_4, θ_5

1. **Solve for** θ_3 using Permutation 3, (p_y)
   Equation form: $A\sin(\theta_3) + B\cos(\theta_3) + C = 0$

$$A = -\frac{5L\left(\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right) + 1\right)\cos\left(\theta_4\right)}{2}$$
$$B = \frac{5\sqrt{2}L\cos\left(\theta_5 + \frac{\pi}{4}\right)}{2}$$
$$C = -5L$$

   Solution (via $t = \tan\frac{\theta_3}{2}$):

$$\theta_3 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

2. **Solve for** θ_7 using Permutation 2, (a_z)
   Equation form: $A\sin(\theta_7) + B\cos(\theta_7) + C = 0$

$$A = 0$$
$$B = 1$$
$$C = \sin\left(\theta_3\right)\sin\left(\theta_5\right)\cos\left(\theta_4\right) - \cos\left(\theta_3\right)\cos\left(\theta_5\right)$$

   Solution (via $t = \tan\frac{\theta_7}{2}$):

$$\theta_7 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

3. **Solve for** θ_6 using Permutation 1, (n_z)
   Equation form: $A\sin(\theta_6) + B\cos(\theta_6) + C = 0$

$$A = 0$$
$$B = 0$$
$$C = \sin\left(\theta_3\right)\cos\left(\theta_4\right) + \sin\left(\theta_5\right)\cos\left(\theta_7\right) + \tilde{\infty}\sin\left(\theta_7\right)\cos\left(\theta_5\right)$$

   Solution (via $t = \tan\frac{\theta_6}{2}$):

$$\theta_6 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

4. **Solve for** θ_2 using Permutation 4, (p_x)
   Equation form: $A\sin(\theta_2) + B\cos(\theta_2) + C = 0$

$$A = 0$$
$$B = -L\cos\left(\theta_3\right)$$
$$C = \frac{L\left(10\sin\left(\theta_3\right) + 4\sin\left(\theta_4\right)\sin\left(\theta_6\right)\cos\left(\theta_7\right) - 4\sin\left(\theta_5\right)\sin\left(\theta_7\right)\cos\left(\theta_4\right) + 5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos\left(\theta_4\right) - 4\cos\left(\theta_3\right) - 4\cos\left(\theta_4\right)\cos\left(\theta_5\right)\cos\left(\theta_6\right)\cos\left(\theta_7\right) + \right.}{2}$$

   Solution (via $t = \tan\frac{\theta_2}{2}$):

$$\theta_2 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

5. **Solve for** θ_1 using Permutation 1, (n_x)
   Equation form: $A\sin(\theta_1) + B\cos(\theta_1) + C = 0$

$$A = \sin(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2)$$
$$B = -\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4)$$
$$C = \sin(\theta_5)\sin(\theta_7) + \cos(\theta_5)\cos(\theta_6)\cos(\theta_7)$$

   Solution (via $t = \tan\frac{\theta_1}{2}$):

$$\theta_1 = 2\arctan\left(\frac{B \pm \sqrt{A^2 + B^2 - C^2}}{A + C}\right)$$

6 Numerical Verification

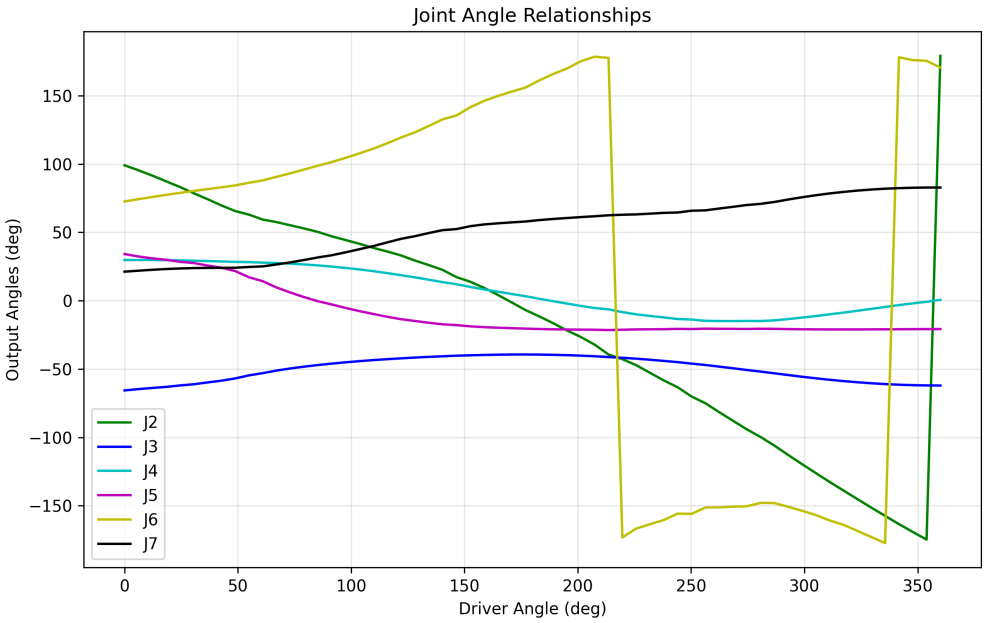


Figure 1: Joint relationships simulation